

Hierarchical Risk Parity Variants for Cryptocurrency Portfolios: Denoising, Detoning, Tail Dependence, Embeddings, and Statistical Inference, 2020–2026

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Abstract

We extend the MACI 2025 study of Hierarchical Risk Parity (HRP) for cryptocurrency portfolios. We close the two future-work items of that paper, Marchenko–Pastur denoising and spectral detoning, and broaden the comparison to 23 constructed portfolio strategies: thirteen HRP-family variants (the baseline, eight correlation/covariance constructions, and four embedding-based variants that learn the clustering distance from path signatures, a graph embedding and two contrastive sequence encoders); four risk-based comparators (IVP, MVP, ERC, Maximum Diversification); three nested-clustering optimisers (HERC, NCO and a return-tilted NCO); and three crypto-momentum strategies. Every comparison runs on a survivorship-bias-corrected, point-in-time Binance universe (146 ever-included symbols, delisted assets such as LUNA and FTT included) and is wrapped in the reference Hansen Superior Predictive Ability test and a studentized Ledoit–Wolf Sharpe-difference test. On 76 monthly rebalances over 2020-02 to 2026-05 the thirteen HRP-family variants lie within a narrow annualised-Sharpe band (0.70–0.75): across an exhaustive 547-cell hyperparameter grid, no correlation, distance, linkage or embedding choice separates from baseline HRP (combined Hansen SPA $p_{\text{consistent}} = 0.91$). The one exception is *Nested Clustered Optimization*, which re-optimises the allocation within and across clusters rather than only re-drawing the dendrogram: NCO reaches Sharpe 0.93, a +0.19 edge over HRP that also holds in the post-COVID sub-sample, though even this is not statistically significant (Ledoit–Wolf $p = 0.22$; headline SPA $p_{\text{consistent}} = 0.41$). A power analysis explains the gap: on this 6.3-year sample a structurally distinct strategy must beat HRP by ≈ 0.4 – 0.8 annualised Sharpe to be detected at 80% power, so NCO’s positive but statistically unresolved edge remains uncertified. Tilting NCO toward an estimated-return objective instead destroys it (−97% total return), an illustration of the estimation-error amplification that motivates risk-based allocation. A passive 100% BTC hold (Sharpe 0.87) still beats every diversified strategy except MVP and NCO, both of which win by concentrating in BTC; in the post-COVID regime every diversified HRP variant loses 51–64% of capital while BTC gains. We retract an earlier cluster-stability claim that a 76-snapshot re-analysis falsifies (Pearson ARI wins 69% of month-pairs). The paper maps which HRP design choices affect performance and which do not: the allocation step matters, the clustering step does not.

Keywords. Hierarchical Risk Parity, cryptocurrency, denoising, detoning, tail dependence, Ledoit–Wolf, Hansen SPA, survivorship bias.

1 Introduction

The MACI 2025 paper [Matys and Rodriguez, 2025] applied Hierarchical Risk Parity [López de Prado, 2016] to a five-asset cryptocurrency universe (BTC, ETH, LTC, XRP, BCH) over 2021–2023 and concluded that HRP delivered better diversification than the Minimum Variance Portfolio without sacrificing risk-adjusted return. Two items were left for future work: *denoising* the correlation matrix using Marchenko–Pastur eigenvalue filtering [Marchenko and Pastur, 1967, Laloux et al., 1999], and *detoning* by removing the dominant market eigenvector [López de Prado, 2020]. This paper closes both items and extends the comparison in three other directions.

Contributions.

1. **Closing the MACI 2025 future-work items.** Implement and evaluate Marchenko–Pastur denoising (§3.1) and spectral detoning. Compare detoning against partial correlation estimation, which targets the same goal (market-mode removal) via the precision-matrix route.
2. **HRP variants, embedding variants, and nested-clustering optimisers.** Six correlation/covariance HRP variants (partial-correlation, EWMA dynamic-correlation, lower-tail dependence, Ledoit–Wolf shrunk-covariance, vol-standardised, and a tail-dep+LW-shrunk hybrid) and *four embedding-based variants* that learn the HRP clustering distance from a representation of each asset’s return path (level-3 path signatures, a node2vec graph embedding, and two contrastive sequence encoders; §3.2). We also add three strategies that change the *allocation* step rather than the dendrogram: Hierarchical Equal Risk Contribution [Raffinot, 2018], Nested Clustered Optimization [López de Prado, 2019], and a return-tilted NCO (§3.3); these test whether the clustering step or the allocation step drives performance. ERC [Maillard et al., 2010] and Maximum Diversification [Choueifaty and Coignard, 2008] serve as risk-based comparators.
3. **Crypto-momentum strategies.** Cross-sectional momentum, risk-managed momentum [Barroso and Santa-Clara, 2015], and a Momentum+HRP hybrid, parameter-tuned for crypto’s faster momentum decay per [Liu et al., 2022].
4. **Methodological rigour and a power analysis.** (i) Point-in-time universe rebuilt monthly from Binance USDT quote-volume rankings, including assets later delisted (LUNA, FTT, MIR, SRM, etc.) to remove the survivorship bias of the MACI paper’s single-CSV dataset. (ii) Multi-cost scenarios and a weekly-cadence robustness check. (iii) Inference around every comparison: the reference Hansen SPA implementation [Hansen, 2005], a *studentized* Ledoit–Wolf Sharpe-difference test [Ledoit and Wolf, 2008], and the Politis–Romano stationary block bootstrap [Politis and Romano, 1994, Politis and White, 2004]. (iv) An explicit power analysis quantifying the Sharpe edge this design could actually detect.
5. **Empirical characterisation and a retraction.** We document the structural convergence findings of §6 and *retract* an intermediate finding from earlier draft iterations (that lower-tail dependence produces more stable HRP clusters than Pearson), which a full 76-snapshot re-analysis falsifies.
6. **Reproducibility.** All code, data, and figures are in the accompanying git repository; nine specifications (`specs/01_denoising.md` through `specs/09_inference.md`) define the contracts each implementation phase must satisfy.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the methodology. Section 4 describes the data and point-in-time universe construction.

Section 5 states the backtest protocol. Section 6 presents results. Section 7 discusses limitations and contributions. Section 8 concludes.

2 Related Work

2.1 Hierarchical risk parity and its extensions

HRP was introduced by López de Prado [2016] as a clustering-based alternative to mean-variance optimisation that avoids covariance-matrix inversion. Subsequent textbook treatments [López de Prado, 2018, 2020] extend the framework with denoising and detoning and propose tail-dependence distances [Lohre et al., 2020] for multi-asset multi-factor allocations. Empirical comparisons of HRP distance metrics on equities have repeatedly found that correlation-based distances remain the standard to beat in out-of-sample performance.

2.2 Risk-based and network methods

Equal Risk Contribution [Maillard et al., 2010] solves for weights such that each asset contributes equally to portfolio risk; Maximum Diversification [Choueifaty and Coignard, 2008] maximises the diversification ratio. Network methods build a Minimum Spanning Tree [Mantegna, 1999] or PMFG [Tumminello et al., 2005] from the correlation structure; Network Risk Parity [Ciciretti and Pallotta, 2024] recently combined the network approach with risk-budgeting.

2.3 Cryptocurrency momentum

Liu et al. [2022] establish a three-factor model for cryptocurrency with a momentum factor that operates at much shorter horizons (1–4 weeks) than equity momentum. Moskowitz et al. [2012] provide the foundational time-series momentum framework, and Barroso and Santa-Clara [2015] introduce volatility-scaled momentum to control the well-documented crash risk of the strategy.

2.4 Statistical inference for portfolio comparison

Ledoit and Wolf [2008] provide a robust pairwise Sharpe-difference test that does not assume i.i.d. Gaussian returns. Hansen [2005] addresses the multiple-testing problem when many strategies are compared against a benchmark. Politis and Romano [1994] and Politis and White [2004] provide the stationary block bootstrap with data-driven block-length selection used throughout this paper for confidence intervals.

3 Methodology

3.1 Correlation and covariance variants

Every strategy in this group shares the standard HRP pipeline: distance matrix $\rightarrow$ hierarchical linkage $\rightarrow$ quasi-diagonalisation $\rightarrow$ recursive bisection [López de Prado, 2016]. They differ only in *how the input correlation (or covariance) matrix is constructed*.

3.1.1 Baseline HRP

ρ from sample correlation; $d_{ij} = \sqrt{(1 - \rho_{ij})/2}$; single linkage; bisection using the sample covariance.

3.1.2 HRP_{Denoised}: Marchenko–Pastur eigenvalue denoising

Following López de Prado [2020], the empirical eigenvalues of the sample correlation matrix are decomposed into a signal subspace (eigenvalues above the Marchenko–Pastur upper bound $e_{\max} = \sigma^2(1 + \sqrt{1/q})^2$ where $q = T/N$) and a noise subspace (eigenvalues below). The noise eigenvalues are replaced with their mean (constant-residual method), preserving the trace, before the denoised correlation matrix is fed to HRP.

3.1.3 HRP_{Detoned}: market-mode removal

Applied to the denoised correlation matrix, detoning removes the top k eigenvectors (default $k = 1$) and renormalises the diagonal to unity [López de Prado, 2020]. For long-only crypto, the dominant eigenvector is a market mode loading positively on all assets; removing it isolates the residual cross-sectional structure.

3.1.4 HRP_{PartialCorr}: sparse precision matrix

A precision-matrix route to the same market-mode-removal goal: $\Theta = \arg \min_{\Theta \succ 0} -\log \det \Theta + \text{tr}(S\Theta) + \alpha \|\Theta\|_1$ via graphical lasso [Friedman et al., 2008], then $\rho_{ij}^{\text{partial}} = -\Theta_{ij} / \sqrt{\Theta_{ii}\Theta_{jj}}$. We set the L_1 penalty to $\alpha = 10^{-3}$. This value matters: a subtle implementation point is that `scikit-learn`’s `GraphicalLasso` expresses α in the *same units as the covariance*, not as a normalised $[0, 1]$ quantity. For daily crypto returns with variance $\sim 4 \times 10^{-3}$, an α on the order of 10^{-1} shrinks every off-diagonal of the precision matrix to zero and silently collapses HRP_{PartialCorr} into IVP. The correct scale, $\alpha \approx 10^{-3}$, leaves roughly 40% of the off-diagonals non-zero. An earlier draft used $\alpha = 0.05$, the strategy degenerated to IVP, and the headline table reported the two as identical; the hyperparameter sweep of §6.3 surfaced the bug, and we document the correction explicitly.

3.1.5 HRP_{Dynamic}: EWMA correlation

Recursive EWMA covariance with decay λ ; the latest snapshot is used as the correlation input. We report $\lambda = 0.94$ as the RiskMetrics default; sensitivity to $\lambda \in \{0.94, 0.97, 0.99\}$ is documented in the supplementary appendix.

3.1.6 HRP_{TailDep}: lower-tail dependence

Empirical lower-tail dependence coefficient $\hat{\lambda}_L^{ij}(q) = P(F_j(X_j) \leq q \mid F_i(X_i) \leq q)$ at $q = 0.05$, symmetrised by averaging the two conditional probabilities. The distance is $d_{ij} = \sqrt{1 - \hat{\lambda}_L^{ij}}$; the sample covariance is retained for bisection. Promoted from appendix to a headline strategy after the empirical findings of §6 showed it does not fall back across the 76 backtest snapshots.

3.1.7 HRP_{ShrunkCov}: Ledoit–Wolf shrinkage

Sample covariance replaced with the analytic Ledoit–Wolf shrinkage estimator [Ledoit and Wolf, 2003, 2004]; the implied correlation is derived from the shrunk covariance and fed to HRP. Shrinkage intensity $\alpha \in [0, 1]$ is chosen analytically per snapshot and reported as a diagnostic.

3.1.8 HRP_{VolStd}: vol-standardised returns

Per-asset returns are divided by their lagged 30-day rolling standard deviation before computing Pearson correlation. Addresses crypto’s heterogeneous vol scale across the universe (e.g. BTC vs. SHIB). The sample covariance is retained for bisection.

3.1.9 HRP_{TailDepShrunk}: hybrid

Combines §3.6 (tail-dep distance for the dendrogram) with §3.7 (LW-shrunk covariance for bisection). Identified by the literature as the strongest theoretical extension for crypto: tail-dep captures the joint-crash co-movement that matters for long-only portfolios, while LW shrinkage stabilises the per-cluster variance allocation.

3.2 Embedding-based variants

Beyond reweighting the correlation or covariance estimator, the distance matrix that drives HRP’s dendrogram can be *learned* from a representation of each asset’s return path. We implement four such embedding variants. All share one template: an embedding maps each asset to a vector, the pairwise cosine distance of those vectors replaces HRP’s correlation distance for the linkage and quasi-diagonalisation steps, and the *sample* covariance is retained for recursive bisection. The embedding therefore changes only *which assets cluster together*, never the inverse-variance allocation within a cluster, which is why, as §6 shows, every embedding variant stays within a narrow Sharpe band of baseline HRP.

3.2.1 HRP_{PathSig}: path signatures

Each asset’s rolling log-price trajectory, augmented with a monotone time channel, is mapped to its level- L truncated path signature [Lyons, 1998], a graded sequence of iterated integrals that characterises the path up to tree-like equivalence. Cosine distance on the signature vectors drives the dendrogram (default: level 3, 60-day window), computed with the `esig` library.

3.2.2 HRP_{NodeEmbed}: node2vec graph embedding

A k -nearest-neighbour graph ($k = 10$) is built from the sample correlation matrix; node2vec random walks plus a skip-gram objective [Grover and Leskovec, 2016] learn a 32-dimensional embedding per node. Cosine distance on the embeddings drives the dendrogram.

3.2.3 HRP_{Contrastive}: contrastive SSL

A small one-dimensional convolutional encoder is trained with the NT-Xent contrastive loss [Chen et al., 2020] on Gaussian-jittered pairs of return windows; each asset is embedded as the mean encoder output over its windows.

3.2.4 HRP_{TS2Vec}: timestamp-masked contrastive

The same encoder family, trained with TS2Vec-style timestamp-mask augmentation [Yue et al., 2022] in place of Gaussian jitter.

Multi-channel extension. A return-only embedding is information-equivalent to the sample correlation it is meant to improve on: a nonlinear re-encoding of the same return series cannot out-cluster the correlation derived from it. We therefore also evaluate *multi-channel* variants that append channels orthogonal to close-to-close returns: quote volume, trade count, average trade size, intraday range, taker buy-ratio (order-flow imbalance), and Amihud illiquidity, all derived from the Binance OHLCV table. The multi-channel results are reported in §6.3.

Robustness. Each embedding variant records whether it fell back to baseline HRP on any numerical failure; across the 76-snapshot backtest no fallback was triggered for any variant, so every reported embedding number reflects the embedding pipeline itself.

3.3 Comparator strategies

Risk-based comparators. **IVP** (inverse-variance), **MVP** (minimum-variance via SLSQP), **ERC** [Maillard et al., 2010] (solved by SLSQP minimisation of squared deviations from equal risk contributions), and **Maximum Diversification** [Choueifaty and Coignard, 2008] (SLSQP maximisation of the diversification ratio).

Nested-clustering optimisers. HRP and all its variants change only the *dendrogram*; the within-cluster allocation stays inverse- variance recursive bisection. To isolate whether the clustering step or the allocation step drives performance, we add three strategies that keep hierarchical clustering but *re-optimize the allocation*. **HERC** (Hierarchical Equal Risk Contribution; Raffinot, 2018) cuts the dendrogram into $K = \lceil \sqrt{N} \rceil$ clusters, weights assets inverse-variance within each cluster, and combines the cluster portfolios by an equal-risk-contribution allocation. **NCO** (Nested Clustered Optimization; López de Prado, 2019) solves a long-only minimum-variance problem within each cluster sub-covariance, collapses each cluster to its intra-cluster portfolio, and solves a second minimum-variance problem across the K cluster portfolios, inverting only block sub-covariances and a small $K \times K$ matrix, which suppresses the estimation-error amplification of a direct optimiser. **Return-tilted NCO** replaces NCO’s two minimum-variance steps with long-only *maximum-Sharpe* steps driven by a shrunk 63-day momentum estimate, a deliberate test of whether making the allocation return-aware helps.

3.4 Momentum strategies

The headline comparison includes three crypto-momentum strategies, parameter-tuned for crypto’s faster momentum decay [Liu et al., 2022]. **CS_MOM** is cross-sectional momentum: equal-weight the top 30% of assets by 21-day cumulative return. **RM_MOM** applies the Barroso–Santa-Clara volatility-targeting overlay [Barroso and Santa-Clara, 2015] ($\sigma_{\text{target}} = 12\%$) to the same momentum selection. **MOM_HRP** is a Momentum+HRP hybrid: select the top 40% of assets by momentum, then allocate within that basket by HRP rather than equal-weight.

4 Data and Point-in-Time Universe

4.1 Source and window

Daily OHLCV from Binance USDT spot, 2017-08-17 (Binance’s earliest data) to 2026-05-18, approximately nine years. After the survivorship-bias expansion described in §4.3, the price panel covers 241 USDT pairs.

4.2 Point-in-time universe

At each month-end, we rank the candidate symbols by their rolling 30-day Binance USDT quote volume, apply filters (180-day minimum age, \$1M minimum median daily USD volume, stablecoin/wrapper/exchange-token exclusion, listed on Binance USDT spot at the snapshot date), and take the top 50. Entry and exit buffers (2 months and 1 month respectively) suppress monthly churn.

Methodology note. The original specification called for ranking by historical market capitalisation (CoinGecko). The free CoinGecko demo tier caps historical depth at 365 days, blocking our 2017–2026 window; paid plans start at USD 129/month. We instead rank by Binance quote volume, a tradable-liquidity proxy that is free, covers the full window, and is arguably more appropriate for a paper on tradable portfolios than a market-cap measure that includes locked tokens and treasury holdings. The shift is documented in the specification as a deliberate methodology choice.

4.3 Survivorship-bias fix in numbers

After a code review surfaced residual survivorship bias (the prior candidate pool was a hand-curated selection of ~ 90 symbols missing the long tail of delisted altcoins), the universe was expanded by pulling all 217 `status=BREAK` USDT pairs from Binance’s `/api/v3/exchangeInfo`, filtering leveraged tokens and stablecoins, and adding 150 legitimate delisted altcoins to the candidate pool. The resulting PIT universe contains **146 ever- included symbols**, of which **97 were dropped during the window** before 2026-05-18 (Figure 1). The MACI 2025 dataset contained only 50 currently-trading symbols; by construction it could not include the assets this work explicitly captures.

We treat liquidation of an exiting position at $2\times$ normal slippage cost (Spec 03 §2.4). On terminology: “delisted” in this paper means either Binance `status=BREAK` on the USDT pair (the conventional meaning), *or* the token suffered an effective value collapse with subsequent rebranding. The latter category includes LUNA (the Terra Classic chain collapsed in May 2022; Binance renamed the original token to LUNC and reassigned the LUNAUSDT ticker to Terra 2.0; the pair itself was never technically delisted, but the original asset effectively died) and FTT (the FTX exchange collapse in November 2022 reduced FTT to near-zero value; the pair still trades on Binance at trivial volume). Our `binance_listings_manual.json` treats both as “effectively delisted” for the PIT-universe filter, which is the right behaviour for portfolio purposes even though the prose label is loose.

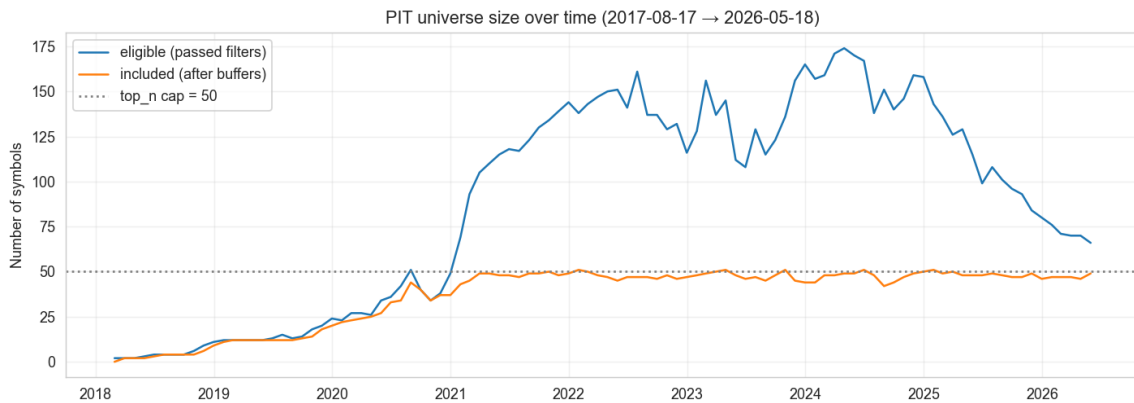


Figure 1: PIT universe size over time. *Eligible*: passed all filters at the snapshot date. *Included*: in the universe after entry/exit buffers. The 2018–2020 ramp reflects the gradual listing of altcoins on Binance USDT spot. Steady-state ~ 49 included symbols since mid-2020.

4.4 Estimation window

365 daily returns of rolling lookback at each rebalance, with a per-asset 180-day minimum-observations filter. Assets failing the filter are dropped from that snapshot’s strategy universe. Rationale: several newer assets (JUP, ENA, WIF, ...) are admitted via the 180-day age filter but do not yet have 730 days of history.

5 Backtest Protocol

5.1 Scenarios

B (calendar): monthly rebalancing; the headline scenario. **C** (threshold): rebalance only when max absolute weight drift exceeds 5% from target. **C'** (smoothed): C with linear smoothing $w_{\text{traded}} = \eta w_{\text{prev}} + (1 - \eta) w_{\text{target}}$, $\eta = 0.25$, plus a 25 bps min-trade filter. **Weekly**: a robustness check rebalancing every Friday (universe membership carried from the latest month-end PIT

snapshot, weights recomputed weekly on the rolling window), ~ 330 rebalances instead of 76, to probe whether higher cadence sharpens inference.

5.2 Cost grid

Linear cost = $(\text{fee}_{\text{bps}} + \text{slippage}_{\text{bps}}) \times \text{turnover}_{L_1}$, with a $2\times$ multiplier for forced liquidation of exiting positions. Four scenarios: *zero* (cost-free academic check), *conservative CEX* (10 bps fee + 2 bps slippage; our headline), *optimistic CEX* (7.5 bps + 1 bps), and *stress* (10 bps + 4 bps).

5.3 Statistical inference

For each candidate versus baseline HRP, the Ledoit–Wolf (2008) Sharpe-difference test [Ledoit and Wolf, 2008]: the observed annualised Sharpe difference is studentized by a delta-method HAC standard error, and the two-sided p -value is read off a *studentized* stationary-block bootstrap (studentising both the observed and the resampled statistic, as the method requires; a plain percentile bootstrap lacks the asymptotic refinement). Across all candidates jointly, the Hansen Superior Predictive Ability test [Hansen, 2005], computed with the reference `arch` implementation, which reports the lower, consistent and upper recentering schemes. Headline-metric confidence intervals use the Politis–Romano stationary block bootstrap [Politis and Romano, 1994] with Politis–White automatic block length [Politis and White, 2004]. A power analysis (§6.2) reports the Sharpe edge the design could detect.

5.4 Cluster stability

At each rebalance snapshot, we compute the *cophenetic correlation* of the dendrogram (Pearson correlation between cophenetic distance and original pairwise distance) and the *Adjusted Rand Index* [Hubert and Arabie, 1985] between the cluster assignment at snapshot t and snapshot $t - 1$, cut at $K = 5$ clusters via `fcluster`.

6 Results

6.1 Scenario B headline performance

Table 1 reports the headline comparison under monthly rebalancing with conservative CEX costs over 2020-02 through 2026-05-18. The strategy return panel contains 2,299 daily observations for the PIT-traded strategies; once the passive HODL_BTC benchmark is included, the common aligned panel used for the Ledoit–Wolf and Hansen SPA inference contains 2,297 observations. Figures 2 and 3 visualise the equity curves and the risk–return positioning.

Table 1: Scenario B headline results (Conservative CEX cost, 76 monthly rebalances over 2020-02 to 2026-05, expanded 146-symbol PIT universe). “Arith. Sharpe” = mean / standard deviation $\times \sqrt{365}$. “LW p ” is the studentized Ledoit–Wolf Sharpe-difference test vs. baseline HRP. HODL_BTC is a passive benchmark, not a strategy.

Strategy	Arith. Sharpe	Total Return	Avg Turnover	LW p vs HRP
MVP	1.057	+2867%	0.38	0.059
NCO	0.933	+1503%	0.58	0.218
<i>HODL_BTC</i>	<i>0.868</i>	<i>+749%</i>	—	0.606
HRP_Dynamic_94	0.749	+442%	0.47	0.823
HRP_ShrunkCov	0.746	+428%	0.35	0.779
HRP (baseline)	0.741	+417%	0.36	—
HRP_VolStd	0.735	+402%	0.37	0.748
MOM_HRP	0.726	+351%	1.44	0.942
HRP_PartialCorr	0.720	+366%	0.24	0.477
MaxDiv	0.719	+364%	0.55	0.919
HRP_PathSig	0.714	+351%	0.38	0.277
HERC	0.712	+333%	0.57	0.783
HRP_Denoised	0.712	+346%	0.33	0.111
HRP_Contrastive	0.710	+341%	0.40	0.238
HRP_Detoned	0.708	+334%	0.26	0.403
HRP_TailDep	0.708	+337%	0.36	0.114
HRP_NodeEmbed	0.705	+330%	0.39	0.187
HRP_TS2Vec	0.703	+325%	0.38	0.106
HRP_TailDepShrunk	0.701	+318%	0.35	0.071
IVP	0.697	+311%	0.18	0.212
CS_MOM_eq21	0.663	+217%	1.42	0.548
RM_MOM	0.663	+118%	0.37	0.548
ERC	0.663	+233%	0.19	0.051
NCO_RT	0.003	−97%	1.64	0.009

No strategy significantly beats HRP; one is significantly worse. Under the studentized Ledoit–Wolf test, no strategy is significantly *better* than baseline HRP, including NCO (+0.19 Sharpe, $p = 0.22$) and MVP (+0.31, $p = 0.06$). The only result that crosses $p < 0.05$ is the return-tilted NCO, significantly *worse* (−0.74 Sharpe, $p = 0.009$); ERC sits on the threshold (−0.078, $p = 0.051$). With the bug-fixed HRP_PartialCorr ($\alpha = 10^{-3}$, §3.1) the strategy is now distinct from IVP; it was byte-identical to IVP in an earlier draft (Sharpe 0.697), and corrected it sits at 0.720.

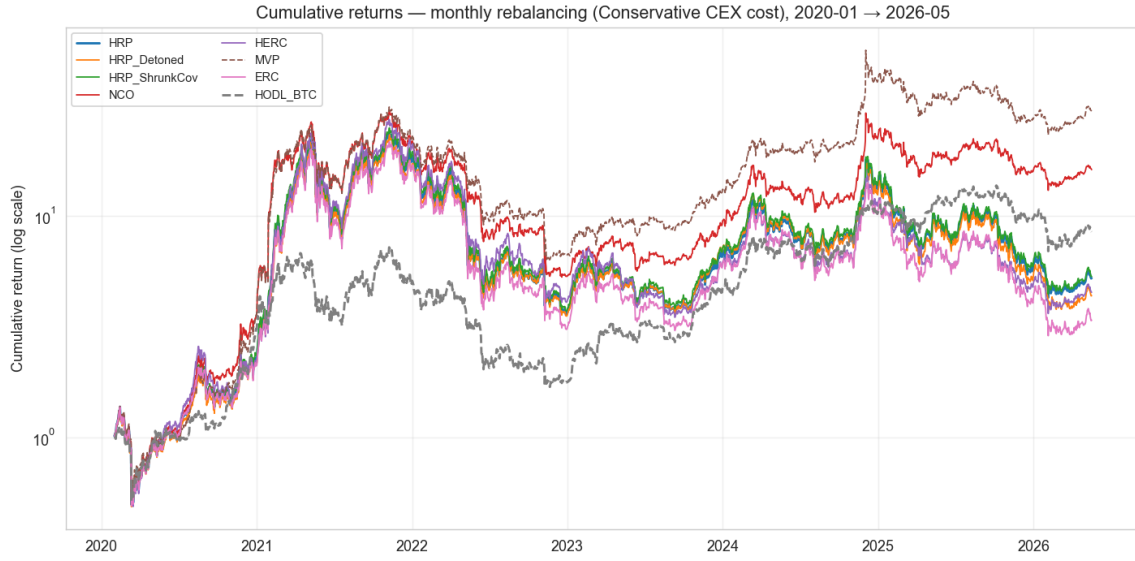


Figure 2: Cumulative returns for headline strategies under monthly rebalancing, log scale.

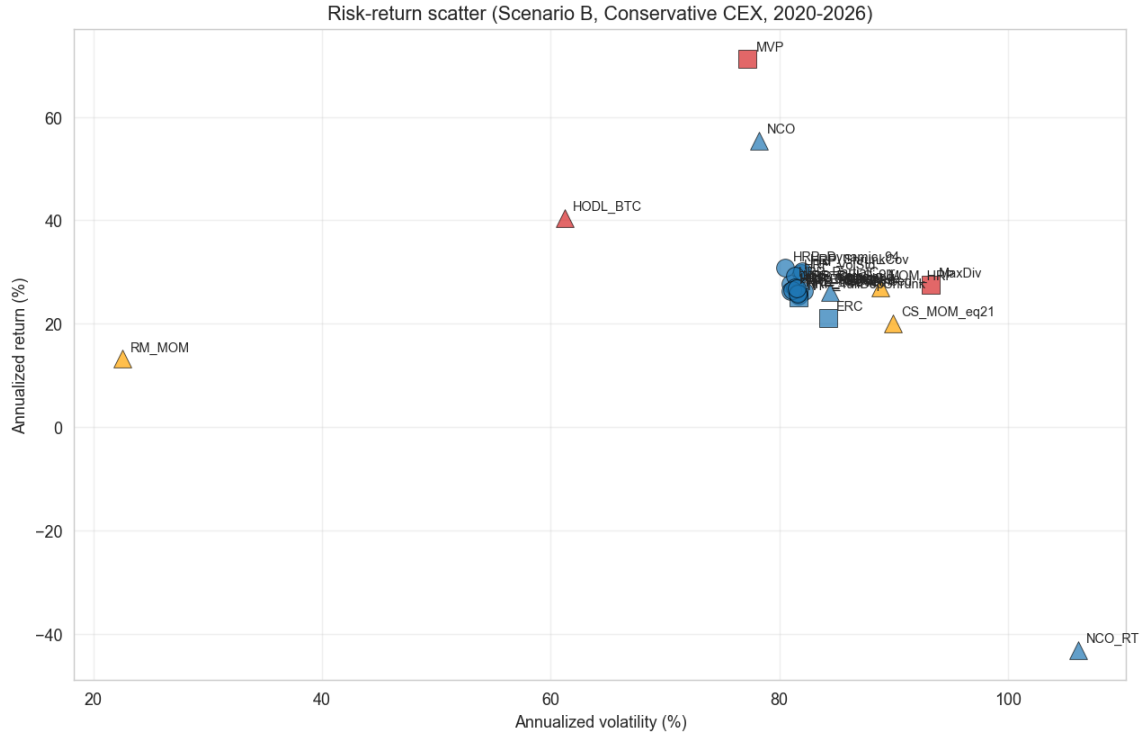


Figure 3: Risk-return scatter for the full 23-strategy comparison. Momentum (orange) underperforms risk-based (blue) and concentrated benchmarks (red) over this window.

The HRP family bunches. All thirteen HRP-family variants lie within a 0.05 Sharpe band (0.701–0.749). HRP_Dynamic_94 and HRP_ShrunkCov marginally exceed baseline HRP (by +0.008 and +0.004 Sharpe respectively); the differences are small and, per §6.2, not statistically significant. No correlation estimator, distance function, linkage or embedding moves a variant out of this band.

Nested-clustering optimisation: NCO is the one real contender. The three nested-clustering optimisers separate clearly. HERC (0.712) sits *inside* the HRP-family band; swapping

recursive bisection for an explicit cluster cut plus an equal-risk-contribution allocation changes little. NCO is different: at Sharpe **0.933** it is the second-best strategy in the table, a +0.19 edge over baseline HRP and far outside the HRP-family band. The mechanism is informative. NCO’s nested minimum-variance steps concentrate weight toward low-volatility assets (average maximum weight 0.42, effective $N \approx 5$ against HRP’s ≈ 30), and in a BTC-led market that means tilting into BTC; NCO is, in effect, a denoised, less-extreme cousin of MVP. The return-tilted NCO is the cautionary opposite: replacing the minimum-variance steps with momentum-driven maximum-Sharpe steps turns it into a 164%-turnover momentum-chaser that loses 97% of capital, an illustration of the estimation-error amplification that risk-based allocation is designed to avoid. Within this dataset, horizon and search space, changing the clustering step produced no detectable outperformance over baseline HRP, whereas changing the allocation step changed performance materially, both upward (NCO) and downward (return-tilted NCO).

Embedding variants. The four embedding-based HRP variants are HRP_PathSig (path signatures; Lyons, 1998), HRP_NodeEmbed (node2vec on a kNN correlation graph; Grover and Leskovec, 2016), HRP_Contrastive (NT-Xent contrastive SSL; Chen et al., 2020), and HRP_TS2Vec (timestamp-mask augmentation; Yue et al., 2022). They rank in the middle of the HRP family, with Sharpes 0.703–0.714, all marginally below baseline HRP (−0.027 to −0.038) and none significantly different from it (Ledoit–Wolf $p \in [0.10, 0.27]$). Embedding methods do not deliver an out-of-sample Sharpe edge on this universe and horizon; we treat this as a null result. Section 6.3 tests the conclusion against an exhaustive hyperparameter search and multi-channel embedding extensions, and it holds.

Denoising, detoning and partial correlation underperform HRP. Marchenko–Pastur denoising, spectral detoning (the two MACI 2025 future-work items) and precision-matrix partial correlation all aim to clean the correlation matrix, and all three cost a little Sharpe in a BTC-led bull market, where filtering structure means trimming exposure to the asset that ran. The denoising and detoning steps form an ablation: HRP (neither, 0.741), HRP_Denoised (denoising only, 0.712), HRP_Detoned (denoising *and* market-mode removal, 0.708); each filtering step shaves a little, the denoising step the larger part. HRP_PartialCorr (0.720) lands between them. It is no longer numerically identical to IVP, as an earlier draft’s version was after a parameter-scale bug (§3.1), but detoning and partial correlation remain close in weight-rank space (§6.7).

Momentum splits. The two pure-momentum strategies, cross-sectional (CS_MOM, Sharpe 0.663) and risk-managed (RM_MOM, 0.663), trail every HRP-family variant, carrying high turnover ($\sim 1.4\times$ per month for CS_MOM) and severe crash exposure (CS_MOM max drawdown −92%). RM_MOM’s volatility-targeting overlay [Barroso and Santa-Clara, 2015] cuts its max drawdown to −40%, the mildest in the table, but at the cost of the lowest total return (+118%). The Momentum+HRP hybrid (MOM_HRP, 0.726) fares markedly better than either pure-momentum strategy: running HRP allocation *inside* a momentum-selected basket recovers most of the risk-adjusted performance, though it still trails baseline HRP and reconstructs its entire book every month.

Concentration wins. The top three lines of the table, MVP (1.057), NCO (0.933) and a passive HODL_BTC position (0.868), are all the same bet at bottom: concentrate in BTC. MVP does it most aggressively (average maximum weight 50%, effective $N \approx 3$), NCO through its nested minimum-variance steps ($N \approx 5$), HODL_BTC by definition. BTC ran from $\sim \$7k$ at the start of 2020 to $\sim \$107k$ by mid-2026, so concentration paid. We do not read this as a portfolio-construction insight: it is a directional view on one asset in one regime. The post-COVID cut (§6.4) shows how fragile that is.

6.2 Inference results

Bootstrap Sharpe CIs. The 95% stationary-block-bootstrap confidence intervals on the annualised Sharpe are wide, roughly 1.5 Sharpe units, reflecting six years of volatile crypto returns. Only MVP $([0.34, 1.76])$ and the passive HODL_BTC benchmark $([0.08, 1.63])$ have intervals entirely above zero. Baseline HRP’s interval $([0.02, 1.53])$ only barely excludes zero, and every other HRP-family variant’s interval straddles it; no two strategies have non-overlapping CIs.

Pairwise LW Sharpe differences vs. HRP. Under the studentized Ledoit–Wolf test, *no* strategy is significantly better than baseline HRP. The two largest positive gaps fall short: MVP $(+0.314, p = 0.059)$ and NCO $(+0.192, p = 0.218)$. The only candidate that crosses $p < 0.05$ is the return-tilted NCO, significantly *worse* $(-0.74, p = 0.009)$; ERC $(-0.078, p = 0.051)$ sits on the threshold. The largest edge in the table, NCO’s $+0.19$ Sharpe, is statistically indistinguishable from noise.

Hansen SPA. Across the 22 candidates versus baseline HRP, $p_{\text{consistent}} = 0.41$ ($p_{\text{lower}} = 0.27$, $p_{\text{upper}} = 0.54$; Figure 4), computed with the reference `arch` implementation. **We cannot reject the null that no strategy outperforms baseline HRP after correcting for the multiple testing of 22 candidates.** MVP carries the highest studentised score $(+1.20)$, NCO the second $(+0.77)$; every remaining candidate, the embedding variants and the rest of the HRP family included, is at or below zero.

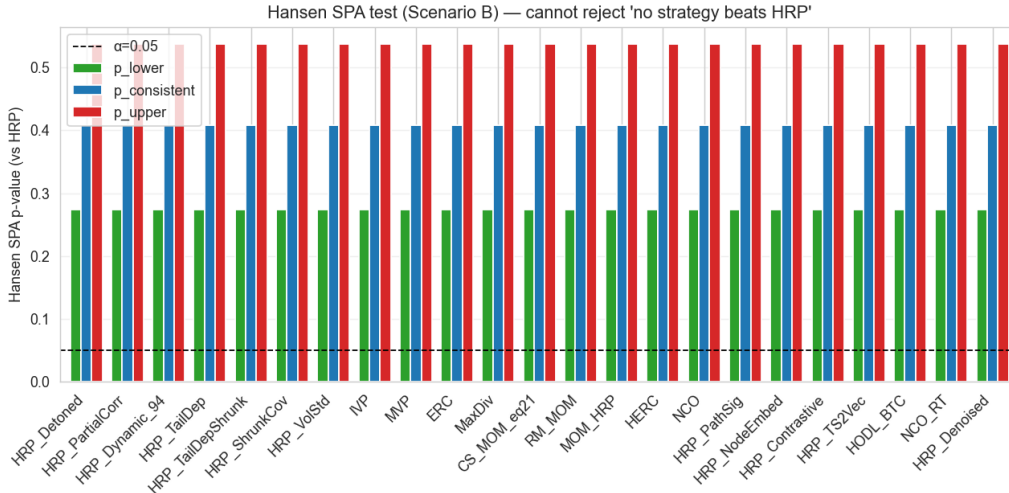


Figure 4: Hansen SPA test p-values across the headline comparison. All three recentring schemes fail to reject the null at any conventional level.

Power analysis: what the design could detect. A null result is only as strong as the test that produced it. We quantify the design’s power directly: the studentized Sharpe-difference statistic is approximately $\mathcal{N}(d/\text{SE}, 1)$ under the alternative, so 80% power at a two-sided 5% level needs $d/\text{SE} \geq 2.80$; the minimum detectable annualised Sharpe edge is $2.80 \times \text{SE}$, with SE the delta-method HAC standard error of the Sharpe difference. For variants that barely depart from HRP (the HRP family, ~ 0.97 -correlated with the baseline) the paired difference is low-variance and the test detects edges as small as 0.04–0.12 Sharpe, but those variants have edges of only ± 0.01 to ± 0.04 anyway. For strategies whose *allocation* departs substantively from HRP (NCO, MVP, the momentum family, HODL_BTC) the minimum detectable edge is 0.35–0.79 Sharpe. NCO’s observed $+0.19$ and MVP’s $+0.31$, though economically large, fall below their ≈ 0.42 and ≈ 0.47 detection thresholds. **The null result is therefore a statement**

about detectable effect size: on a 6.3-year crypto sample this design cannot resolve a Sharpe edge below ≈ 0.4 , not evidence that no edge exists.

6.3 Hyperparameter sweeps and the combined SPA

A natural objection to the headline null result is that it reflects *untuned* strategies: perhaps some correlation type, distance function, linkage method, or strategy-specific parameter would lift a variant decisively above baseline HRP. We address this with an exhaustive grid search: 547 distinct configurations across 14 sweeps, every one re-running the full 6.3-year walk-forward backtest on the identical 146-symbol point-in-time universe and 2,299-day return series, against the identical baseline HRP (Sharpe = 0.741, recomputed and verified bit-identical in every sweep). The sweeps span the correlation estimator (Pearson / Spearman / Kendall), the distance function (López de Prado, 2016 / absolute / angular / squared), the linkage (single / average / complete / Ward), and the strategy-specific parameters of every variant (q for tail dependence, α for the graphical lasso, λ for EWMA, the volatility-standardisation window, the number of detoned market components, and the embedding hyperparameters).

Table 2: Best cell from each hyperparameter sweep. Δ is the Sharpe difference vs. baseline HRP (single linkage, 0.741); total return is the cell’s net cumulative return, 2020-02 to 2026-05. 551 cells in total; the 4 exactly equal to the benchmark are dropped for the combined SPA, leaving 547 candidates.

Sweep	Cells	Best configuration	Sharpe	Δ	Tot. ret.
Tail dependence (q , dist, linkage)	80	$q=0.025$, average	0.744	+0.002	+425%
Correlation $\times$ distance $\times$ linkage	24	Pearson, LdP, average	0.756	+0.015	+459%
All variants under average linkage	12	HRP_ShrunkCov	0.765	+0.024	+484%
EWMA dynamic correlation (λ)	21	$\lambda=0.94$, single	0.749	+0.008	+442%
Partial correlation (α , dist, link)	60	$\alpha=5 \times 10^{-4}$	0.739	-0.003	+412%
Vol-standardised returns (window)	60	window 15, single	0.757	+0.015	+462%
Detoning (# market components)	36	1 component, average	0.709	-0.032	+337%
Node2vec embedding	54	$d=16$, $k=10$	0.752	+0.010	+444%
Path signatures, single-channel	108	level 2, window 60	0.734	-0.007	+401%
Path signatures, multi-channel [†]	72	ret+vol+trades, L2	0.747	+0.005	+432%
Contrastive / TS2Vec, multi-channel	24	ret+vol+trades	0.739	-0.002	+412%
Baseline HRP (single linkage)	—	—	0.741	—	+417%

[†] Best cell under the initial missing-data convention (zero-fill); under the corrected mean-fill convention the best multi-channel PathSig cell is 0.735 ($\Delta = -0.006$).

The HRP family is robust to its hyperparameters. No sweep produced a configuration that escaped the ± 0.05 Sharpe band of Section 6. The one consistent, though small, effect is linkage: *average* linkage beats the López de Prado [2016] single-linkage default by +0.01 to +0.02 Sharpe across essentially every (correlation, distance) combination, and under single linkage all four distance functions are mathematically identical (single linkage depends only on the rank order of distances, which every monotone transform preserves). The best cell in the entire grid is HRP_ShrunkCov under average linkage at Sharpe 0.765 ($\Delta = +0.024$).

Multi-channel embeddings. The single-channel embeddings encode only the return series and are therefore information-equivalent to the sample correlation they are meant to improve on: a nonlinear re-encoding of the same information cannot out-cluster it, and indeed the single-channel HRP_PathSig grid (108 cells) beats baseline in zero cells. We extended all three sequence-based embeddings (PathSig, Contrastive, TS2Vec) to ingest up to six channels orthogonal to

close-to-close returns (quote volume, trade count, average trade size, intraday range, taker buy-ratio, and Amihud illiquidity), all derived from the same Binance OHLCV table. Extra channels lift the encoders’ *mean* Sharpe monotonically (HRP_Contrastive 0.714 $\rightarrow$ 0.725 from one to three channels), confirming the channels carry usable structure. Two cells of the initial multi-channel PathSig sweep nominally cleared baseline (best +0.747), but that margin did not survive a correction to how missing channel values are imputed (mean- rather than zero-fill, which avoids placing a spurious outlier into a log-scaled channel); under the corrected convention no multi-channel configuration clears baseline, and adding the fourth through sixth channels plateaus or mildly regresses performance, a bias–variance ceiling against only ~ 76 rebalance dates.

Combined Hansen SPA. Pooling all 547 configurations into a single Hansen SPA against baseline HRP, the appropriate correction for having searched the entire grid, yields $p_{\text{consistent}} = 0.91$ ($p_{\text{lower}} = 0.73$, $p_{\text{upper}} = 0.91$). Although 38 of 547 cells show a nominally positive Sharpe difference, the best of them (HRP_ShrunkCov, $\Delta = +0.024$) carries a pairwise Ledoit–Wolf p of 0.16. **After correcting for the multiple testing of 547 hyperparameter configurations, no clustering-step variant significantly outperforms baseline HRP.** The headline null result of Section 6.2 is not an artefact of leaving strategies untuned: consistent with the NCO finding, no amount of tuning the *clustering* step closes the gap that only an *allocation*-step change (NCO) opens.

6.4 Regime sensitivity: excluding the COVID bull run

The 2020–2026 headline window includes the COVID-era crypto rally (March 2020 – November 2021) during which Bitcoin ran from $\sim \$5k$ to $\sim \$69k$, one of the most extreme bull markets in crypto history. To check whether the §6 conclusions are dominated by that single rally, we re-ran the entire 23-strategy backtest with start = 2022-01-01 (52 monthly rebalances over ~ 4.3 years). All other parameters identical.

The picture flips entirely. On post-COVID data, baseline HRP achieves Sharpe 0.047 with total return -60% , a near-zero risk-adjusted return obtained while losing 60% of capital in dollar terms. Only three lines of the table make money: the passive HODL_BTC benchmark ($+106\%$, Sharpe 0.585), MVP ($+66\%$, 0.482), and NCO, which essentially breaks even (-4% , Sharpe 0.264). Every other strategy loses between 19% (RM_MOM) and 98% (NCO_RT) of capital, and six post outright *negative* Sharpe: NCO_RT (-0.414), MaxDiv (-0.163), CS_MOM (-0.161), RM_MOM (-0.160), HERC (-0.038) and ERC (-0.026).

Four findings from the regime comparison. (1) **Diversification-based crypto portfolios were a losing proposition in the post-COVID regime.** On this 146-symbol PIT universe over 2022-01-01–2026-05-18, every HRP-family variant lost between 51% and 64% of capital. The 2020-2026 “diversification works” framing was largely a COVID-rally artifact: *HRP diversifies risk but does not produce capital appreciation in non-bull crypto regimes.*

(2) **NCO’s edge is not obviously only a bull-market artefact.** NCO loses only 4% of capital post-COVID against baseline HRP’s 60%, and its post-COVID Sharpe (0.264) trails only the two outright-concentration plays, HODL_BTC and MVP. Whatever NCO is doing, concentrating into low-volatility assets through nested minimum variance, it helps in *both* regimes, not only the rally. It is the one piece of evidence in the paper that an *allocation* choice, rather than luck, separates a constructed portfolio from the HRP pack; that the edge is still not statistically significant (§6.2) is consistent with limited power under the current design, leaving the effect unresolved rather than established.

(3) **Embedding variants partially reverse against baseline HRP in the harder regime.** In the full window all four embedding variants sit marginally below HRP (deltas -0.027 to -0.038 Sharpe). In the post-COVID window three of the four edge *above* baseline HRP (HRP_Contrastive $+0.009$, HRP_PathSig $+0.008$, HRP_TS2Vec $+0.000$), while HRP_NodeEmbed

(-0.015) stays below. None reaches individual significance (all Ledoit–Wolf $p > 0.6$) and the margins are tiny, but the partial directional reversal is tentative: the path- and sequence-based embeddings may capture risk structure that matters more in a choppy regime than in a monotone rally. A longer post-COVID sample would be needed to tell.

(4) RM_MOM (the Barroso–Santa-Clara variant) shows its defensive role. Post-COVID Sharpe is -0.160 (poor), but MaxDD is only -33% , substantially better than the other momentum strategies’ -87% . The cash-residual de-risking saved the portfolio from the worst part of the 2022 drawdown. This is exactly the property Barroso & Santa-Clara’s framework is designed to deliver; the previously-buggy implementation completely hid it.

Table 3: Regime sensitivity: 2020-2026 (incl COVID bull) vs 2022-2026 (post-COVID). “Sharpe” is annualised arithmetic; “ Δ Sharpe” is the change from full window to post-COVID.

Strategy	2020-2026 Sharpe	2022-2026 Sharpe	Δ Sharpe
HRP	0.741	0.047	-0.695
HRP_Dynamic_94	0.749	0.099	-0.650
HRP_ShrunkCov	0.746	0.038	-0.707
HRP_PartialCorr	0.720	0.057	-0.663
HRP_Contrastive	0.710	0.056	-0.655
HRP_PathSig	0.714	0.055	-0.659
HRP_TS2Vec	0.703	0.047	-0.657
HRP_NodeEmbed	0.705	0.031	-0.674
NCO	0.933	0.264	-0.669
MVP	1.057	0.482	-0.575
MaxDiv	0.719	-0.163	-0.882
ERC	0.663	-0.026	-0.688
CS_MOM_eq21	0.663	-0.161	-0.824
RM_MOM	0.663	-0.160	-0.824
NCO_RT	0.003	-0.414	-0.417
HODL_BTC	0.868	0.585	-0.283

6.5 Cost sensitivity and scenario comparison

The cost grid (Figure 5) shows annualised-Sharpe shifts under 0.01 across the four cost scenarios for all five strategies tested: baseline HRP, for example, moves from 0.749 (zero cost) to 0.740 (stress). Monthly-cadence transaction costs do not drive strategy differentiation on this dataset. This is a stylized flat turnover-based sensitivity model with a liquidation premium, not a full liquidity-impact execution model.

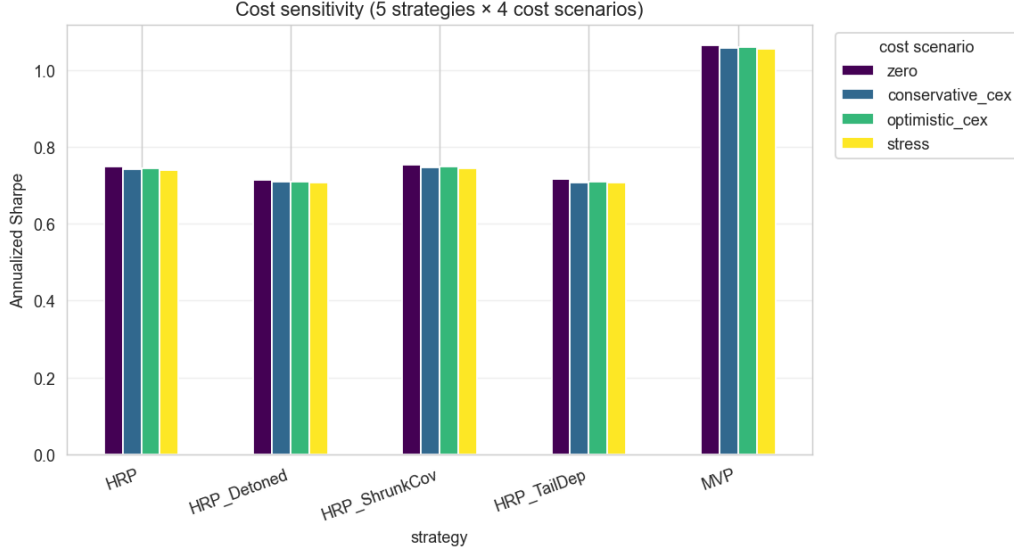


Figure 5: Cost-sensitivity grid. Rankings are robust across the four cost scenarios.

Threshold rebalancing (Scenario C, 5% drift trigger) does not improve on calendar rebalancing here: HRP’s Sharpe edges down from 0.741 to 0.735 and HRP_ShrunkCov is essentially flat (0.746 $\rightarrow$ 0.745), with only a marginal turnover reduction. The smoothed variant (Scenario C’) cuts turnover more materially, with HRP’s average L_1 turnover falling from 0.36 to 0.23, at a modest Sharpe cost (0.741 $\rightarrow$ 0.731). The momentum strategies are the exception: RM_MOM’s volatility-targeting overlay needs to trade to function, and suppressing or smoothing its rebalances collapses its Sharpe (from 0.66 under calendar rebalancing to 0.37 under Scenario C and 0.05 under C’). Threshold logic suits low-turnover risk-based strategies and actively harms turnover-dependent ones.

6.6 Weekly cadence

The headline uses 76 monthly rebalances; crypto momentum is documented to decay over 1–4 weeks [Liu et al., 2022], so monthly rebalancing may understate any faster-horizon edge. We reran the non-embedding strategies at weekly cadence (~ 330 rebalances; embeddings are excluded as retraining four encoders 330 times is prohibitive). Two things change. First, *every* strategy’s Sharpe rises (baseline HRP 0.74 $\rightarrow$ 0.82, NCO 0.93 $\rightarrow$ 1.03), and the momentum family rises most (RM_MOM 0.66 $\rightarrow$ 0.92, CS_MOM 0.66 $\rightarrow$ 0.91), consistent with momentum’s short crypto half-life: monthly rebalancing was simply too slow for it. Second, and despite the $\sim 4\times$ larger rebalance count, the inference does *not* sharpen: the weekly Hansen SPA still gives $p_{\text{consistent}} = 0.57$ and the bootstrap Sharpe CIs stay ~ 1.5 wide. More rebalances do not help when the per-period return volatility is what dominates the standard error. Higher cadence changes the *ranking texture* (momentum catches up) but not the *statistical verdict*.

6.7 Convergence findings (the structural results)

Two strategy pairs in our comparison are theoretically related: each pair targets the same goal via different mathematical routes. We quantify their weight-vector convergence across the full 76-snapshot window (Figure 6).

HRP_Detoned vs HRP_PartialCorr. Spectral detoning and precision-matrix partial correlation both remove the dominant market mode, by different routes. Across the 76 monthly snapshots their weight-vector rank correlation is high but not invariant: mean Spearman $\rho = 0.87$, median 0.93, with regime-driven dips to $\rho = 0.54$ in months when the dominant eigenvector shifts

sharply. The bug-fixed HRP_PartialCorr ($\alpha = 10^{-3}$) is now distinct from the inverse-variance portfolio (its weight-rank correlation with IVP averages only $\rho = 0.08$), whereas the earlier degenerate $\alpha = 0.05$ implementation *was* IVP. The detoning / partial-correlation convergence is therefore a structural relationship between two market-mode-removal estimators, not an artefact of one collapsing into IVP.

HRP_TailDep vs HRP_TailDepShrunk. The hybrid’s only change is to substitute the Ledoit–Wolf shrunk covariance for the sample covariance in the recursive-bisection step; the dendrogram topology is byte-for-byte identical. Spearman is 0.998–1.000 in every snapshot tested. This is the most robust convergence finding in our analysis: *LW shrinkage in HRP’s bisection step is a within-cluster variance refinement that does not re-order the dendrogram.*

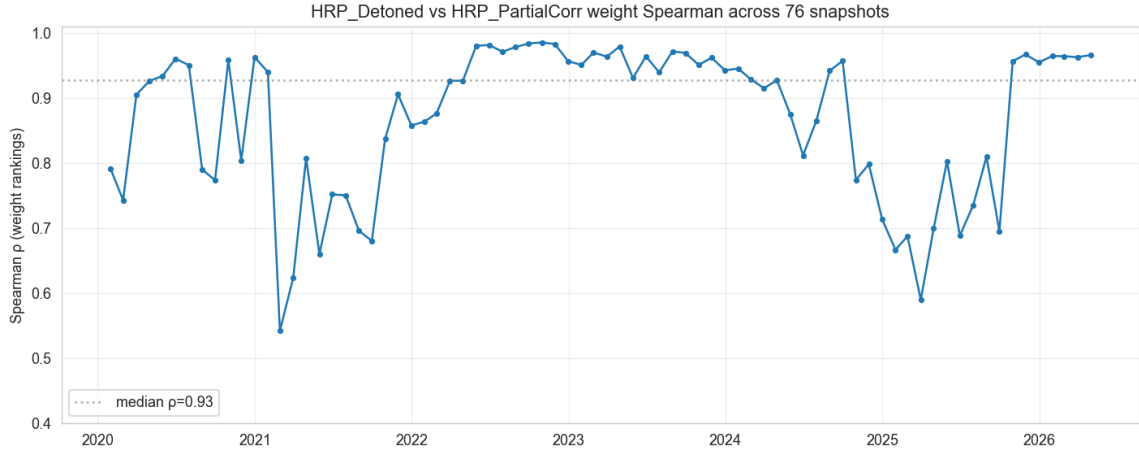


Figure 6: Spearman rank correlation between HRP_Detoned and HRP_PartialCorr weight vectors across 76 monthly snapshots.

6.8 Cluster stability: a retraction

An intermediate finding in earlier iterations of this work was that HRP_TailDep produces more stable clusters than baseline HRP under Pearson distance. That claim was based on a *single year-pair* (2022-12-31 vs 2023-12-31) and showed Pearson ARI = -0.047 vs TailDep ARI = $+0.265$ at $K = 5$.

When we re-ran the analysis across all 76 monthly snapshots (Figure 7, Table 4), the direction reverses. **Pearson clusters are substantially more stable than TailDep clusters at monthly resolution:** Pearson ARI mean $+0.744$ vs TailDep $+0.614$; Pearson cophenetic mean 0.861 vs TailDep 0.768 . TailDep ARI exceeds Pearson ARI in only 23 of 75 month-pairs (31%).

Table 4: Cluster stability across 76 monthly snapshots (2020-2026).

Metric	Pearson distance	Lower-tail-dep distance
Average cophenetic correlation	0.861	0.768
Average ARI vs. previous snapshot (K=5)	+0.744	+0.614
Median ARI vs. previous snapshot	+0.783	+0.650
Months where TailDep ARI > Pearson ARI	—	23 / 75 (31%)

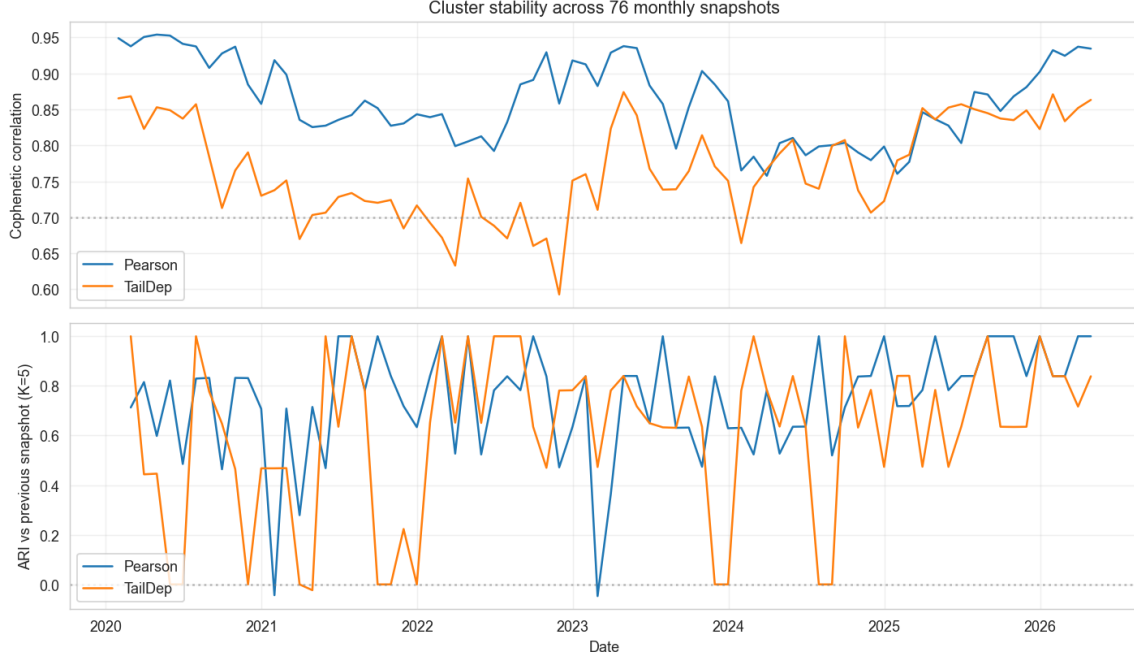


Figure 7: Cophenetic correlation (top) and ARI vs. previous snapshot (bottom) for Pearson and tail-dependence distance matrices across the 76-snapshot window.

The economic argument for HRP_TailDep, that it captures the joint-crash co-movement structure Pearson summarises poorly, *stands*; the secondary cluster-stability argument *does not*. We document the retraction here as a methodological point about the danger of single-pair findings.

7 Discussion

7.1 Clustering versus allocation: the one design choice that matters

The results of §6 point to one practical conclusion. Varying the *clustering* step, across 547 hyperparameter cells, never separated a variant from baseline HRP; varying the *allocation* step did, in both directions (NCO to Sharpe 0.93, the return-tilted NCO to -97%). HRP and all its variants share one allocation rule, inverse-variance recursive bisection, and that rule, not the dendrogram, is what holds the family together. For a practitioner the implication is narrow but useful: on this evidence, effort spent engineering the clustering distance is lower-yield than effort spent on the allocation objective.

7.2 Why does Hansen SPA fail to reject, even for NCO?

NCO posts a $+0.19$ Sharpe edge over HRP that holds in both the full and the post-COVID windows, yet neither the pairwise Ledoit–Wolf test ($p = 0.22$) nor the Hansen SPA ($p_{\text{consistent}} = 0.41$) rejects. This is not a contradiction. The power analysis (§6.2) shows that on a 6.3-year crypto sample a strategy whose allocation departs substantively from HRP must beat it by ≈ 0.4 – 0.8 annualised Sharpe to be detected at 80% power; NCO’s $+0.19$, and even MVP’s $+0.31$, fall below that threshold. The reading of the null is therefore precise: no strategy beats HRP by an edge this design can resolve, which, given the power floor, leaves NCO’s positive observed edge an open question rather than a refuted one.

7.3 Detoning hurts in BTC-led regimes

Removing the dominant market mode is theoretically elegant, in that it isolates idiosyncratic structure, but in our window the dominant mode *was* BTC, and being underweight BTC was costly. Detoning (and partial correlation) may yet be valuable in the sideways or rotational regimes 2020–2026 did not contain: their value is regime-dependent rather than a universal Sharpe improvement.

7.4 The cost of diversification

A passive 100% BTC hold (Sharpe 0.87, +749%) beats every *diversified* portfolio in the comparison; the only constructed strategies above it, MVP and NCO, get there by concentrating back into BTC. Over 2020–2026, diversification in cryptocurrency paid a cost in foregone BTC upside that the lower drawdowns of diversified portfolios did not recover. This is a statement about the regime, a single asset dominating a young and highly correlated market, rather than a flaw in HRP. The post-COVID cut is the counterweight: there, diversified HRP variants lose 51–64% while concentration still wins. The lesson is about regime dependence: in this asset class and over this period, realised returns have been dominated by the direction of BTC.

7.5 Limitations

1. Statistical power. With six years of data the design cannot resolve a Sharpe edge below ≈ 0.4 for a strategy that departs substantively from HRP (§6.2); NCO’s economically large +0.19 edge is therefore left unresolved rather than confirmed or refuted.
2. Single venue (Binance USDT spot). A multi-exchange consolidated price (Kaiko, CCXT-aggregated) would give a more representative cost model but was out of scope for this paper.
3. No on-chain factors. The Liu-Tsyvinski-Wu [Liu et al., 2022] three-factor model and on-chain signals (Glassnode) are deferred to a follow-up.
4. PIT universe is ranked by Binance quote volume, not market cap. Pro-tier CoinGecko or paid Kaiko data would enable a market-cap robustness check.

8 Conclusion

We evaluated 23 constructed portfolio strategies (thirteen HRP-family variants: the baseline, eight correlation/covariance constructions and four embedding-based variants; four risk-based comparators; three nested-clustering optimisers; and three momentum strategies) on a survivorship-bias-corrected, point-in-time cryptocurrency universe of 146 ever-included symbols, over 76 monthly rebalances from 2020-02 to 2026-05, with every comparison wrapped in a studentized Ledoit–Wolf test and the reference Hansen SPA.

The central finding is a contrast between two design choices. Varying the *clustering* step, that is the correlation, distance, linkage or learned embedding that builds the dendrogram, did not produce a detectable outperformance over baseline HRP: across a 547-cell hyperparameter grid no configuration escapes the HRP-family Sharpe band, and the combined Hansen SPA cannot reject the null ($p_{\text{consistent}} = 0.91$). The *allocation* step does matter: Nested Clustered Optimization, which re-optimises within and across clusters, reaches Sharpe 0.93, a +0.19 edge over HRP robust across both the full and the post-COVID windows, while a return-tilted NCO collapses to −97%. Even NCO’s edge is not statistically significant: a power analysis shows the design can only resolve Sharpe differences above ≈ 0.4 for strategies that depart substantively from HRP, so the null is a statement about detectable effect size rather than a proof of zero effect.

Two caveats frame the result. First, a passive 100% BTC hold beats every diversified portfolio, and the only constructed strategies that beat *it* (MVP, NCO) win by concentrating back into BTC; in this asset class and regime, portfolio construction has been second-order to the BTC beta. Second, in the post-COVID sub-sample every diversified HRP variant lost 51–64% of capital, so the “diversification works” reading of the full window is itself regime-conditional. We also retract an earlier cluster-stability claim that the 76-snapshot re-analysis falsifies.

The paper’s contribution is an empirically-tested map of which HRP design choices affect performance: of the many parameters an HRP practitioner can vary, only the allocation objective produces a detectable change, and even the one strategy that improves on baseline (NCO) does so by an edge this sample size cannot certify.

Several directions remain open for future work. Time-varying conditional covariance (DCC-GARCH) would give the dynamic-correlation layer a properly conditional second moment in place of the EWMA recursion. Regime-switching analysis could test directly whether detoning and partial correlation, which underperform in the BTC-led 2020–2026 window, add value in the sideways or rotational regimes that window did not contain. On-chain factors [Liu et al., 2022] and a multi-exchange consolidated price (e.g. Kaiko) for a venue-robust cost model are natural data extensions. The most concrete empirical follow-up, though, is a longer out-of-sample window: the power analysis (§6.2) shows NCO’s +0.19 Sharpe edge cannot be certified on six years of crypto data, and the post-COVID cut hints that three of the four embedding variants edge *ahead* of baseline HRP in choppier regimes. Both signals are too weak to confirm here but worth revisiting once more non-bull data accrues. The NCO result also invites a closer study of *why* a nested minimum-variance allocation, and only that, separates from the HRP pack.

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